Exp 7: Integrating a Zoho Creator application with another Zoho service using connections and APIs.

Register Number: 241401047

1. Aim

To integrate the "Loan Application & Pre-Screening Management" application built in Zoho Creator with Zoho CRM, enabling automatic creation of Lead records for submitted loan applicants, thereby demonstrating the use of Zoho Creator's Connection and REST API framework for inter-service data exchange.

2. Procedure

2.1 Problem Statement & Integration Scope

The bank's loan officers currently work only within Zoho Creator, while the bank's sales and relationship-management team tracks prospective customers in Zoho CRM. Loan applicants captured through the Loan_Application_Form are not visible to the CRM-based follow-up team, requiring manual re-entry of applicant details. This experiment eliminates that manual step by establishing a direct integration from the Loan Application report to Zoho CRM's Leads module, triggered on demand by the loan officer.

- **Source Application:** Loan Application & Pre-Screening Management (Zoho Creator)
- **Target Service:** Zoho CRM (Leads module)
- **Integration Trigger:** Officer-initiated custom action on the Loan_Application_Form_Report
- **Data Transferred:** Applicant Name, Email Address, Contact Number, Loan Type

2.2 Target Service Setup

A Zoho CRM organization ("CRM Teamspace") was provisioned under the same Zoho account used for Creator, granting access to the standard Leads, Contacts, Accounts, and Deals modules required for this integration.

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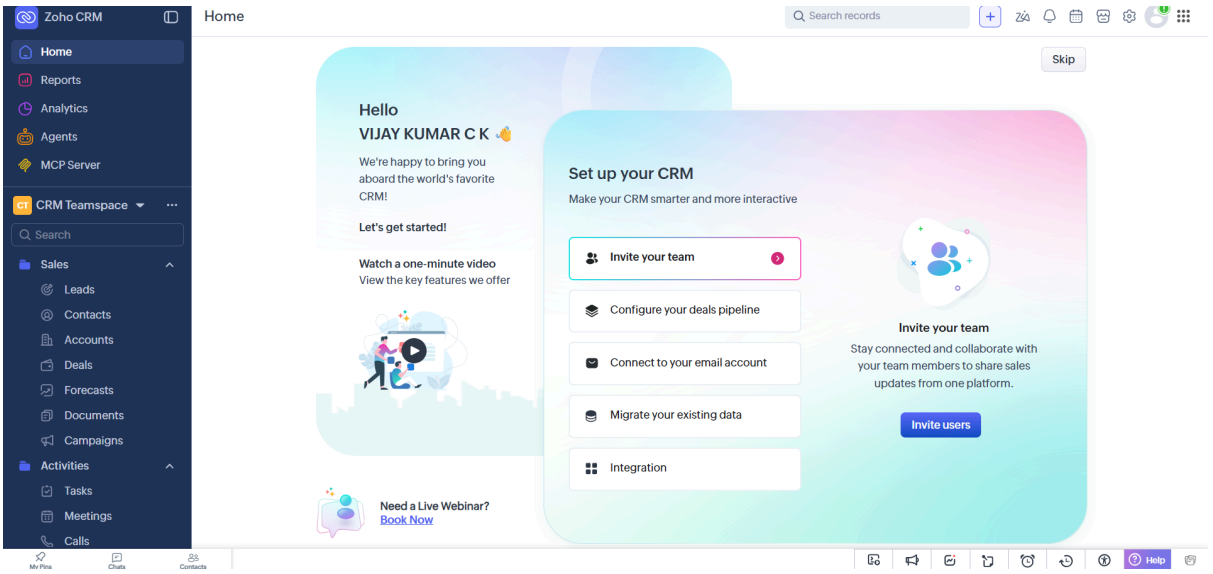


Figure 1: Zoho CRM Home Dashboard Confirming Successful Account Provisioning

2.3 Attempting Native Connection Setup

Zoho Creator provides a built-in Connections microservice that manages OAuth authorization with other Zoho services without manual token handling. This option was found not under the individual application's Settings panel, but under the account-level Microservices dashboard (Creator Home → Microservices → Connections).

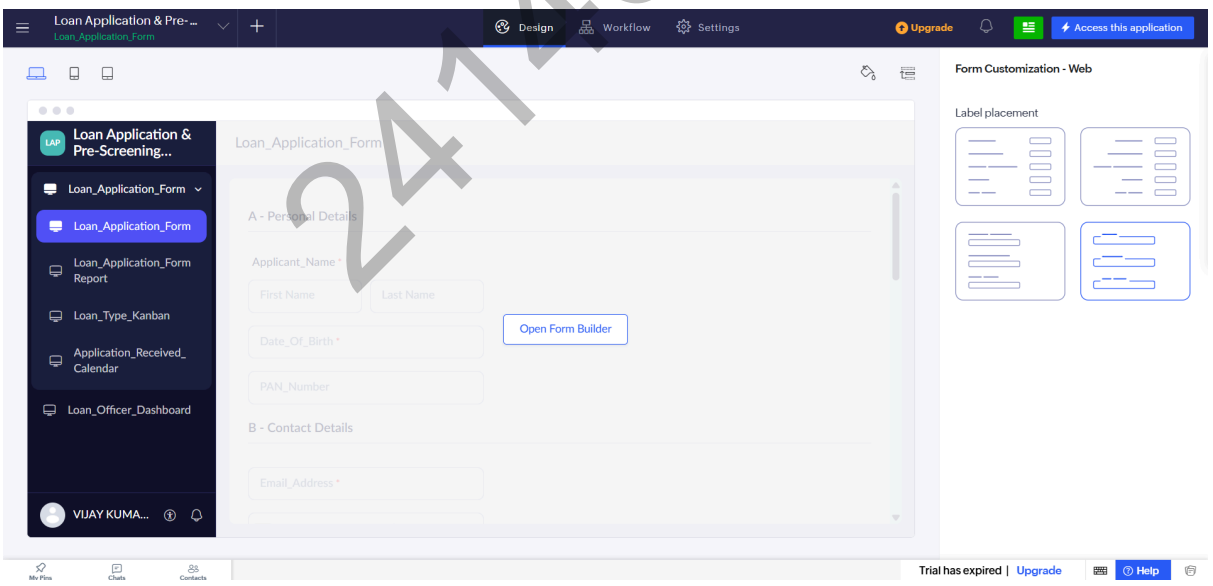


Figure 2: Application-Level Settings Panel (Permissions, Personalization, Developer Tools) — No Connections Option Present Here.

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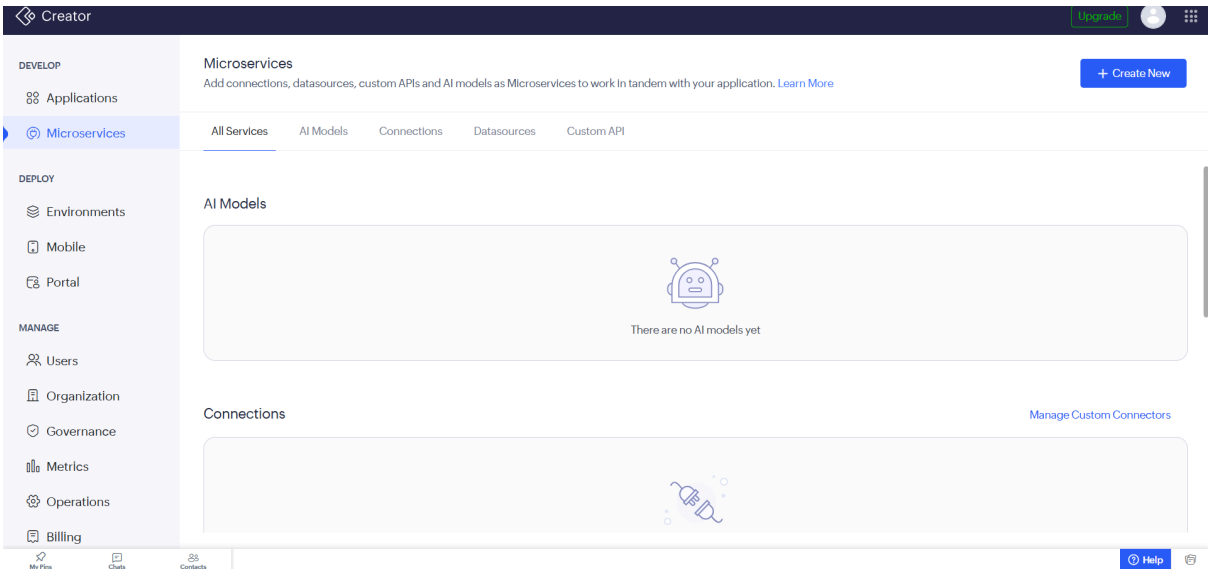


Figure 3: Microservices Dashboard Showing AI Models and Connections Sections

A new connection was configured with the following parameters:

- **Connection Name:** CRM_Connection
- **Version:** Zoho CRM V1
- **Authorization Access:** Administrators only
- **Connection Mode:** Quick Connect (Zoho Credential)
- **Access Scope:** All Actions (all CRM modules, including Leads, Contacts, Accounts, Deals)

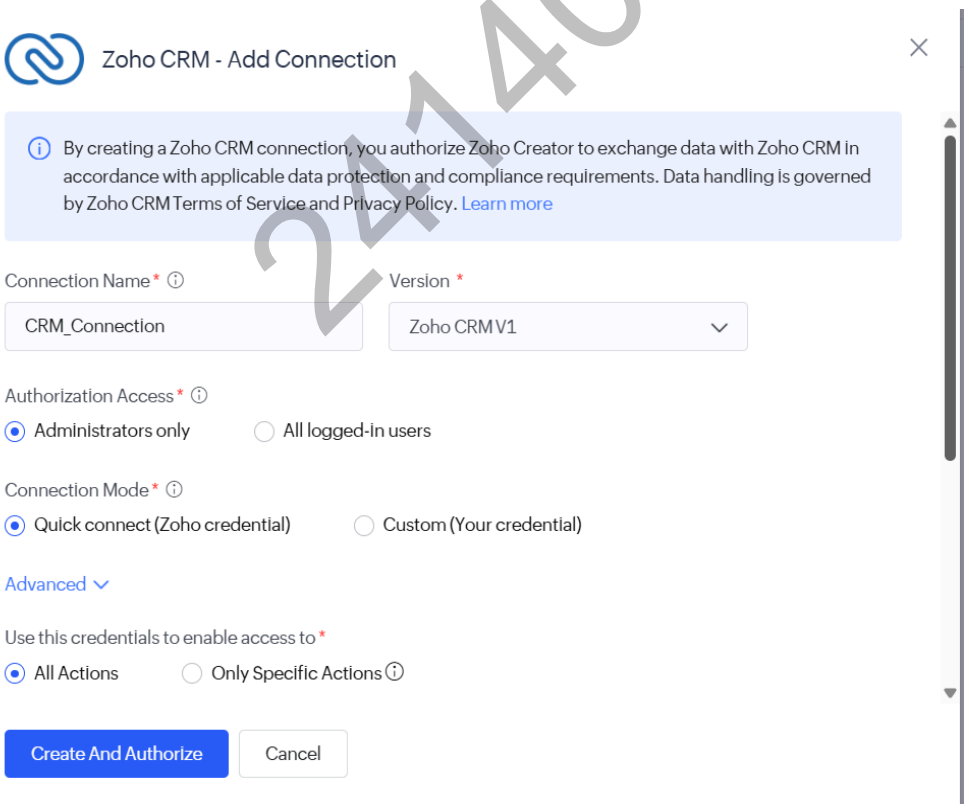
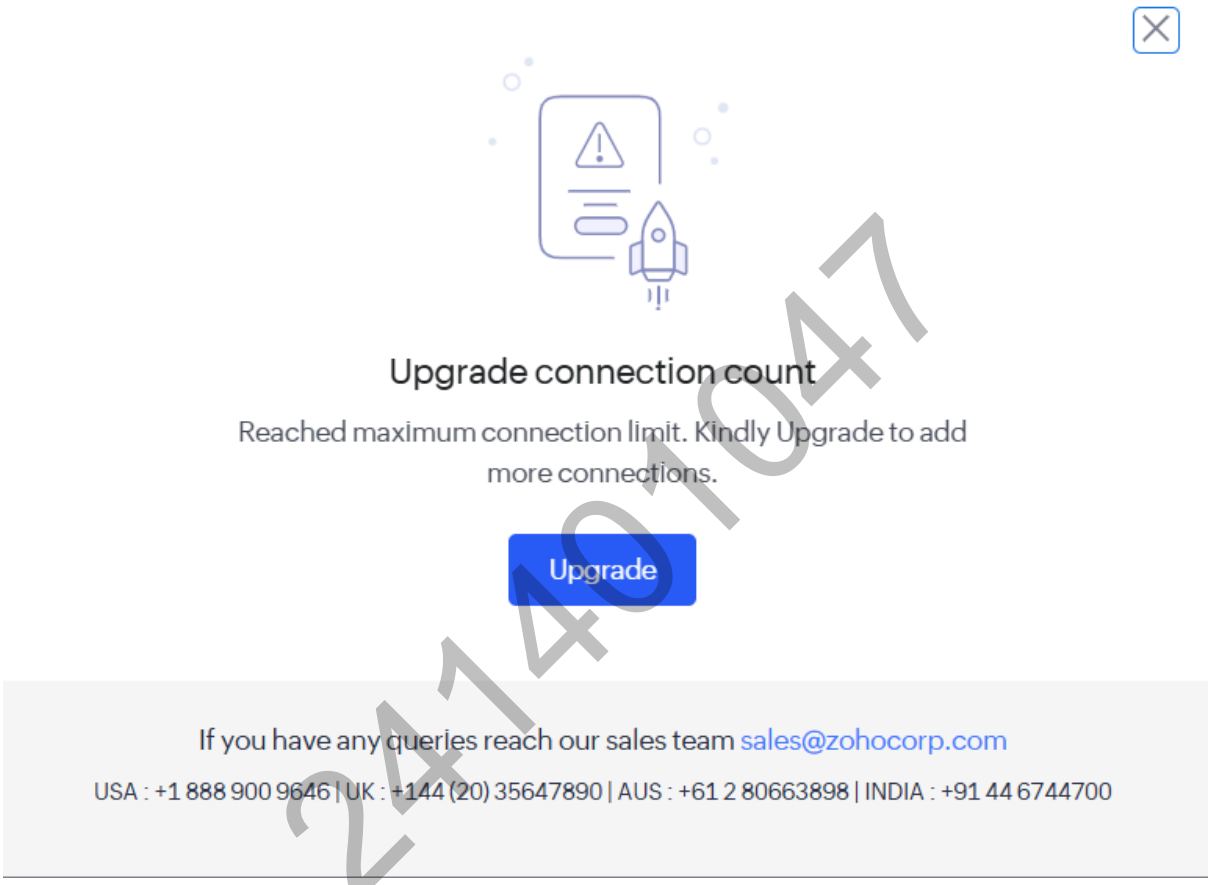


Figure 4: Zoho CRM – Add Connection Configuration Screen

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2.4 Limitation Encountered: Trial Plan Restriction

On clicking "Create And Authorize," Zoho Creator returned an "Upgrade connection count" error, indicating that the account's expired trial plan permits zero active Connections — a plan-level restriction rather than a configuration error, since this was the very first connection attempted on the account.



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Figure 5: "Upgrade Connection Count" Error Returned by Zoho Creator

2.5 Alternative Integration Approach: OAuth Self Client with invokeurl

To work around the plan restriction while still fulfilling the objective of integrating via "connections and APIs," a manual OAuth 2.0 flow was implemented using the Zoho API Console's Self Client feature, combined with Deluge's invokeurl task to call the Zoho CRM REST API directly.

1. A Self Client was created at api-console.zoho.com, generating a Client ID and Client Secret.

2. An authorization grant code was generated with scope ZohoCRM.modules.ALL.
3. The grant code was exchanged for an access token using a POST request via the curl command-line utility, since the token endpoint does not accept GET requests:

```shell
curl -X POST "https://accounts.zoho.in/oauth/v2/token" \
  -d "grant_type=authorization_code" \
  -d "client_id=<CLIENT_ID>" \
  -d "client_secret=<CLIENT_SECRET>" \
  -d "redirect_uri=https://www.zoho.com" \
  -d "code=<GRANT_CODE>"
```

4. The resulting access_token was embedded in a Deluge script attached to a custom action button on the Loan_Application_Form_Report.

2.6 Deluge Script: Push Applicant as Lead to Zoho CRM

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```javascript
headerMap = Map();
headerMap.put("Authorization","Zoho-oauthtoken <ACCESS_TOKEN>");

leadMap = Map();
leadMap.put("Last_Name", input.Applicant_Name);
leadMap.put("Email", input.Email_Address);
leadMap.put("Phone", input.Contact_Number);
leadMap.put("Lead_Source","Zoho Creator - Loan Application");
leadMap.put("Description","Loan Type: " + input.Loan_Type);

dataList = List();
dataList.add(leadMap);

paramMap = Map();
paramMap.put("data", dataList);

response = invokeurl
[
      url: "https://www.zohoapis.in/crm/v2/Leads"
      type: POST
      parameters: paramMap.toString()
      headers: headerMap
];
```

```
info response;
```

This script was attached via Loan_Application_Form_Report → Quick View → Actions → Add New Action Item, configured to execute "At once" without a confirmation prompt, and made accessible from the record-level "..." options menu.

Table 1: API Field Mapping between Loan_Application_Form and Zoho CRM Leads Module

Creator Field	CRM Lead Field	Purpose
Applicant_Name	Last_Name	Mandatory CRM identifier field
Email_Address	Email	Applicant contact email
Contact_Number	Phone	Applicant contact number
Loan_Type	Description	Loan context note
(static value)	Lead_Source	Tags record origin as "Zoho Creator - Loan Application"

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3. Output

: log messages

{"data":[{"code":"SUCCESS","details":{"Modified_Time":"2026-10-05T23:12:26+05:30","Modified_By":{"name":"VIJAY KUMAR C K","id":"[redacted]"Created_Time":"2026-10-05T23:12:26+05:30","id":"[redacted]"Created_By":{"name":"VIJAY KUMAR C K","id":"[redacted]}},"message":"record added","status":"success"}]}

Figure 6: Deluge invokeurl Success Response Log (status: success, message: record added)

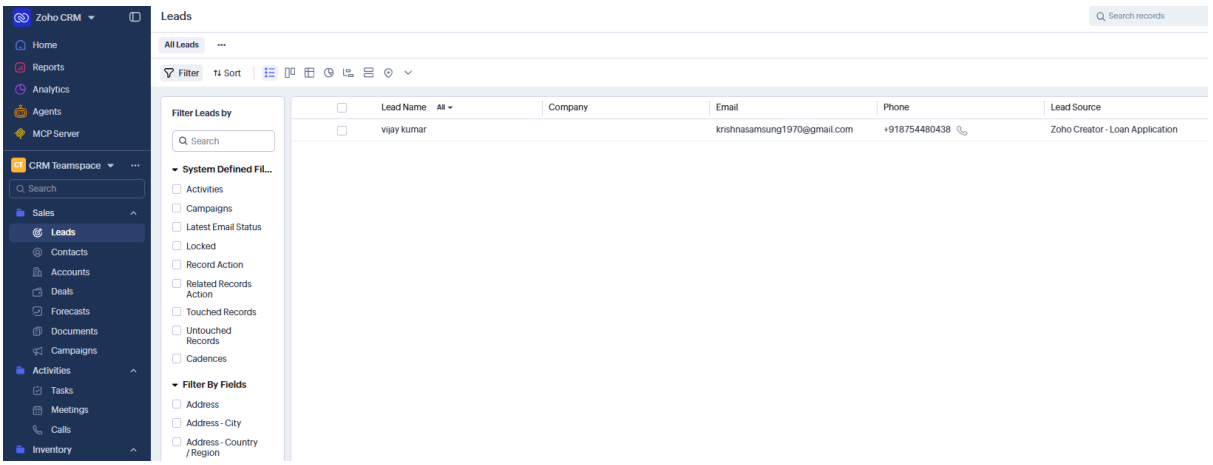


Figure 7: Zoho CRM Leads Module Showing the Newly Created Lead Record with Correct Field Mapping

4. Result

The Loan Application & Pre-Screening Management application was successfully integrated with Zoho CRM. Although the native Connections microservice could not be authorized due to a trial-plan restriction, an equivalent and fully functional integration was achieved using OAuth 2.0 Self Client credentials and Deluge's invokeurl task, successfully creating a Lead record in Zoho CRM directly from a loan officer-triggered action within Zoho Creator.

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